



REVIEW

Are seed oils really bad for you?

REFERENCES 80 verified	STYLE Scientific	TOKENS ≈16.1k	VERIFICATION Crossref	COMPILED 27 Aug 2026
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ABSTRACT

For most people, ordinary consumption of modern, nonhydrogenated seed oils is not supported as a general health hazard. The strongest human evidence indicates that oils rich in unsaturated fatty acids lower low-density-lipoprotein cholesterol when they replace butter, lard, palm oil, or other saturated fats; randomized feeding studies also show neutral or modestly favorable effects on glucose–insulin regulation. Prospective cohorts generally associate higher dietary or circulating linoleic acid with lower cardiovascular, diabetes, and mortality risk, although these observations cannot exclude residual confounding. Two recovered-data analyses of older randomized trials found no mortality benefit, and one found harm, after very large linoleic-acid interventions. Those trials deserve weight but do not cleanly represent current oils or diets because of incomplete data, secondary-prevention or institutional populations, short exposure, and historical margarines containing industrial trans fats. Human trials do not show that increasing linoleic acid generally raises inflammatory markers. Evidence for cancer is mostly null, with a recent colorectal-cancer signal that conflicts with tissue-biomarker analyses. Repeated deep-frying and oil reuse generate oxidation products and should be distinguished from using fresh oil in ordinary cooking. The practical conclusion is replacement-specific: seed oils are usually preferable to saturated fats, but adding any oil raises energy intake, and the healthfulness of fried or ultra-processed foods is not established by the oil alone.

01 Introduction

“Seed oils” is a culinary and social-media category, not a single biological exposure. It commonly includes soybean, corn, sunflower, safflower, cottonseed, grapeseed, rice-bran, sesame, and canola oils, yet their proportions of saturated, monounsaturated, omega-6, and omega-3 fatty acids differ substantially. Even umbrella reviews of named vegetable oils find different lipid, glycaemic, and anthropometric effects across oils, with most associations supported by low or very-low certainty evidence.^{1,2} The central question is therefore not whether a heterogeneous category is “toxic,” but what happens when a particular oil replaces a particular food under a particular cooking condition.

The main disputed constituent is linoleic acid, an essential omega-6 polyunsaturated fatty acid. The harm hypothesis proposes that higher linoleic acid increases arachidonic-acid-derived inflammatory mediators or oxidation products, thereby promoting cardiovascular disease, diabetes, or cancer. The competing hypothesis is that replacing saturated fat with linoleic-acid-rich oils lowers atherogenic lipoproteins and may improve insulin regulation without materially increasing inflammation. Human evidence supports the second hypothesis more consistently, but the old cardiovas-

cular outcome trials and high-temperature reuse literature define important limits.

02 The exposure and the comparator

The same tablespoon of oil can represent opposite dietary changes. Replacing butter or lard with soybean, canola, corn, safflower, or sunflower oil generally exchanges saturated fat for unsaturated fat. Adding oil to an unchanged diet adds energy. Using oil to deep-fry refined starch produces a food whose energy density, sodium, texture, and processing may dominate risk. Reusing an oil repeatedly at high temperature creates an oxidation exposure not represented by fresh-oil feeding trials. Reviews that collapse these interventions obscure the causal contrast.^{1,2,3,4}

The chemical source is also an unreliable shortcut. Coconut and palm are plant oils but are relatively saturated; canola is a seed oil but is rich in monounsaturated fat and contains alpha-linolenic acid; traditional sunflower and safflower oils are linoleic-acid-rich, whereas high-oleic varieties are not. In randomized comparisons, coconut and palm oils raise low-density lipoprotein cholesterol relative to less-saturated vegetable oils, while canola oil generally lowers it.^{5,6,7,8,9,10} Thus, “plant,” “seed,” “refined,” and “natural” do not predict the same metabolic effect.

EXPOSURE	REPLACEMENT	BEST-SUPPORTED INFERENCE	MAIN CAVEAT
Fresh PUFA oil	Butter/lard	LDL falls	Events less certain
Seed oil	Carbohydrate	Often neutral-favorable	Carbohydrate quality varies
Added oil	Nothing	Energy rises	Dose matters
Reused frying oil	Fresh fat	Oxidation rises	Human trials sparse

The replacement principle is the throughline for the clinical evidence.

03 Blood lipids provide the clearest randomized signal

Randomized feeding trials consistently show that unsaturated oils lower total and LDL cholesterol when they replace saturated fats. A network meta-analysis of 54 trials found that safflower, sunflower, rapeseed, flaxseed, corn, olive, soybean, palm, and coconut oils lowered LDL cholesterol by approximately 0.23–0.42 mmol/L relative to butter, with the largest rankings for safflower and rapeseed oils.¹¹ Separate syntheses found higher LDL cholesterol with coconut oil than with non-tropical oils, higher LDL cholesterol with palm oil than with less-saturated vegetable oils, and lower LDL cholesterol with canola than with saturated-fat comparators.^{5,6,7,8,9}

Head-to-head trials support the same direction. In a four-week trial of 96 adults, butter raised LDL cholesterol more than coconut or olive oil.¹² In a multicentre controlled-feeding study, diets low in saturated fat and differing in unsaturated-oil profiles improved cholesterol efflux and related markers.^{13,14} A six-week crossover trial found lipid improvement across three low-saturated-fat, high-unsaturated-fat diets, whether the unsaturated fat came from walnuts or matched oils.¹⁵ A 2024 trial likewise found that dietary linoleic and alpha-linolenic acids modestly altered atherogenic lipoprotein traits rather than producing a uniform “seed-oil” effect.¹⁶

These studies mostly measure risk factors rather than heart attacks. That is a real limitation, but LDL cholesterol is not merely an associative marker; lowering LDL through multiple interventions reduces atherosclerotic risk. The randomized oil data therefore establish a favorable intermediate mechanism, not a complete proof of long-term benefit for every oil, dose, and population.

Clinical guidance that recommends unsaturated replacement rests on this convergence rather than on the botanical origin of an oil. A meta-analysis of randomized trials found fewer coronary events when polyunsaturated fat replaced saturated fat, although the contributing trials were heterogeneous and largely historical.¹⁷ The replacement framing matters: it does not recommend maximizing oil intake, and it does not imply that every processed food made with soybean or sunflower oil is health-promoting. It specifies a nutrient exchange inside a dietary pattern.

04 Cardiovascular events favor unsaturated replacement, with residual uncertainty

Meta-analyses of randomized dietary trials generally find fewer cardiovascular events when saturated fat is reduced or PUFA is increased, but little or no clear effect on total mortality. The 2020 Cochrane review found that sustained saturated-fat reduction reduced combined cardiovascular events, with replacement by PUFA or carbohydrate appearing more favorable than no replacement.¹⁸ A separate Cochrane synthesis concluded that increasing total PUFA probably slightly reduces coronary and cardiovascular events but has little or no effect on all-cause or cardiovascular mortality.¹⁹ When restricted to omega-6 interventions, another Cochrane review found no clear reduction in most cardiovascular outcomes, although myocardial infarction may decrease.²⁰ The apparent disagreement is mostly one of outcome, intervention definition, and certainty—not evidence of a large hidden hazard.

Prospective evidence is more consistently favorable. A pooled biomarker analysis of 30 cohorts, 68,659 participants, and 15,198 cardiovascular events associated higher circulating linoleic acid with lower total cardiovascular disease (hazard ratio per interquintile range 0.93, 95% confidence interval 0.88–0.99), cardiovascular mortality (0.78, 0.70–0.85), and ischaemic stroke (0.88, 0.79–0.98).²¹ A newer individual-level analysis of three large cohorts and an updated meta-analysis also found no cardiovascular penalty from higher circulating linoleic acid.²² Other cohorts report inverse, null, or subgroup-specific associations, but few show a consistent increase in risk.^{23,24,25,26,27}

Dietary analyses similarly associate replacing saturated or animal fat with plant fat or PUFA with lower risk. Findings span PREDIMED, US cohorts, people with diabetes, and recent large studies of plant oils versus butter.^{28,29,30,31,32,33,34,35} These estimates remain observational and model-dependent. Nevertheless, the convergence of LDL lowering, randomized event syntheses, and mostly favorable cohorts argues against a substantial cardiovascular toxicity from ordinary linoleic-acid-rich oils.

The size of the observational association is modest and depends on exposure measurement. A meta-analysis of prospective cohorts associated the highest linoleic-acid intake with 15% fewer coronary events and 21% fewer coronary deaths; modeling a 5%-of-energy replacement of saturated fat yielded smaller estimates of 9% and 13%, respectively.³⁶ A mortality synthesis found relative risks near 0.87–0.89 for the highest versus lowest intake or biomarker categories

across total, cardiovascular, and cancer mortality.³⁷ Such precision should not be mistaken for randomized certainty, but the direction is incompatible with a common, large toxic effect.

05 The old randomized trials are important but not decisive

The strongest contrary evidence comes from recovered data from two trials conducted between 1966 and 1973. In the Sydney Diet Heart Study, 458 men with a recent coronary event were assigned to replace saturated fats with safflower oil and safflower-oil margarine. Mortality was higher in the intervention group: 17.6% versus 11.8% for all causes, hazard ratio 1.62 (95% CI 1.00–2.64).³⁸ In the Minnesota Coronary Experiment, 9,423 institutionalized participants were randomized; the intervention lowered cholesterol substantially but did not reduce mortality, and only 2,355 participants had at least one year of controlled-diet exposure.³⁹

These results cannot be dismissed because they are inconvenient. They show that lowering serum cholesterol with a very high-linoleic-acid dietary package did not guarantee lower mortality in those settings. Yet they also do not

identify modern liquid seed oils as the cause. The Sydney trial had incomplete recovered data, enrolled only men after coronary events, and used period margarines whose trans-fat content was incompletely characterized. Minnesota had variable and often short inpatient exposure, incomplete event-level raw data, only 149 recovered autopsies, and control foods that included trans-fat-rich shortenings and margarines. Both interventions produced dietary patterns and linoleic-acid doses unlike modest household replacement today.^{38,39}

Other historical trials, including the Los Angeles and Oslo diet-heart studies, differed in setting, co-interventions, and oil composition, making a pure linoleic-acid effect difficult to isolate.^{40,41} Modern syntheses that include historical trials reach different conclusions depending on whether multifactorial interventions, omega-3-containing oils, and poorly characterized margarines are pooled.^{42,20,19} The appropriate inference is uncertainty about mortality effects of large, isolated omega-6 interventions—not evidence that customary seed-oil use is broadly harmful.

EVIDENCE STREAM	DIRECTION	CONFIDENCE	WHY
LDL feeding trials	Favorable	High	Direct replacement
Event meta-analyses	Small benefit/null	Moderate	Old heterogeneous trials
Biomarker cohorts	Favorable/null	Moderate	Residual confounding
Historical recovered trials	Null/harm	Low-moderate	Incomplete, era-specific
Inflammation trials	Null	Moderate	Mostly short markers

06 Diabetes, insulin regulation, and body composition

Randomized feeding evidence does not support insulin resistance as a general consequence of plant PUFA. A meta-analysis of 102 controlled feeding trials involving 4,220 adults found that replacing 5% of energy from saturated fat with PUFA improved glucose, glycated haemoglobin, C-peptide, and insulin-resistance measures; replacing carbohydrate with PUFA lowered glycated haemoglobin by 0.11 percentage points.⁴³ A meta-analysis of 13 plant-PUFA feeding trials found small reductions in fasting insulin and the homeostatic model assessment of insulin resistance, without a material change in fasting glucose.⁴⁴ Trials in established type 2 diabetes are more heterogeneous, partly because many compare one unsaturated oil with another.^{45,46}

Prospective studies are consistent with modest benefit but cannot prove it. Across 31 cohorts, higher dietary linoleic acid was associated with 6% lower diabetes risk, while each standard-deviation increment in tissue or blood linoleic acid was associated with 15% lower risk.⁴⁷ An individual-level analysis of 39,740 adults from 20 cohorts found a stronger biomarker association: risk ratio per interquintile range 0.65 (95% CI 0.60–0.72).⁴⁸ However, plasma linoleic acid only weakly reflects reported intake and may also mark en-

dogenous metabolism; in post-infarction patients, plasma but not dietary linoleic acid predicted lower diabetes risk.⁴⁹ This distinction prevents a biomarker association from becoming a dietary causal claim.

Body-composition trials also contradict a simple obesogenic mechanism. In a seven-week overfeeding trial, sunflower-oil PUFA and palm-oil saturated fat produced similar weight gain, but saturated fat produced more liver and visceral fat, whereas PUFA produced more lean tissue.⁵⁰ In an older controlled trial, replacing saturated fat with PUFA improved insulin sensitivity and abdominal-fat distribution.⁵¹ In a 12-month trial among adults with prediabetes or diabetes, a plant-based high-PUFA diet reduced liver fat and LDL cholesterol versus usual care, although a healthy Nordic pattern improved more secondary outcomes.⁵² The repeated pattern is that energy surplus drives weight gain, while fat type can alter where energy is stored and which metabolic markers change.

07 The inflammation mechanism is plausible but not observed as general harm

Linoleic acid can be converted through fatty-acid-desaturase pathways to arachidonic acid, and arachidonic acid is a substrate for both pro-inflammatory and inflammation-resolving mediators. That biochemical fact establishes pos-

sibility, not the net effect of eating an oil. Human intervention syntheses find virtually no evidence that increasing dietary linoleic acid raises C-reactive protein, cytokines, adhesion molecules, or other standard inflammatory markers in healthy adults.^{53,54,55}

Direct arachidonic-acid and omega-6 interventions also fail to show a uniform inflammatory response. In resistance-trained men, arachidonic-acid supplementation changed plasma and muscle fatty-acid composition without increasing basal systemic or intramuscular inflammation.⁵⁶ Moderate n-6 supplementation did not impair inflammatory-cell function in healthy adults.⁵⁷ FADS1 genotype modifies conversion and some metabolic responses to linoleic- and gamma-linolenic-acid-containing oils, but genotype studies have not established a subgroup in whom customary seed-oil intake causes clinical inflammatory disease.^{58,59}

The precursor-product model is also quantitatively constrained. A systematic review of increased arachidonic-acid intake found that human tissue levels and downstream responses do not rise in the simple proportional manner implied by the “omega-6 equals inflammation” narrative.⁶⁰ Both inflammatory and resolving mediators arise from this pathway, and their net production depends on enzymes, competing substrates, cell state, and dose. The absence of a consistent inflammatory-marker response in controlled trials is therefore more informative for ordinary intake than pathway diagrams alone.

A short randomized trial confirms that linoleic acid is incorporated into blood lipids and mitochondrial cardiolipin.⁶¹ Such mechanistic changes are real, but the study did not show that the direction of the biomarker change predicts disease. The mechanistic evidence therefore supports biological responsiveness, not a generalized inflammatory-toxicity claim.

08 Oxidation and repeated frying are a separate question

Polyunsaturated oils are more susceptible to oxidation than saturated fats under heat, oxygen, light, and time. Repeated high-temperature heating can generate polar compounds, aldehydes, oxidized fatty acids, and polymers; reviews of repeatedly heated oils describe adverse vascular and metabolic effects, but much of the causal evidence comes from animals or unusually intense exposures.^{4,3} This evidence supports avoiding repeated reuse and prolonged overheating. It does not establish that a fresh bottle of canola or soybean oil used for ordinary sautéing causes the same exposure.

Human outcome studies usually measure fried-food intake rather than oil oxidation directly. A meta-analysis associated the highest fried-food intake with major cardiovascular events (summary [relative risk](#) 1.28, 95% CI 1.15–1.43) but not cardiovascular or all-cause mortality; heterogeneity was high and oil type, reuse, food, sodium, and restaurant pattern were inseparable.⁶² Another meta-analysis found associations with overweight and hypertension but no clear

prospective association with type 2 diabetes.⁶³ In two US cohorts, frequent fried-food intake was associated with diabetes and coronary disease, with substantial attenuation after accounting for body weight and cardiometabolic intermediates.⁶⁴ Fried-food findings are reasons to limit frequent deep-fried food, not evidence that uncooked or lightly heated seed oil is intrinsically harmful.

09 Cancer evidence is mostly null, with site-specific uncertainty

The claim that omega-6 oils broadly cause cancer is not supported by prospective human evidence. An early review and meta-analysis found no consistent cancer association with linoleic-acid intake.⁶⁵ A later synthesis of 70 prospective reports found dietary omega-6 intake unrelated to overall cancer risk (RR 1.02, 95% CI 0.99–1.05), while higher blood omega-6 levels were modestly inversely associated.⁶⁶ Breast-cancer meta-analysis was null, and prospective breast-cancer cohorts have produced mixed or null results.^{67,68,69,70} A prostate-cancer dose-response analysis found no association with dietary linoleic acid and a modest inverse association with tissue levels.⁷¹

The main recent contrary finding is colorectal cancer. A 2025 meta-analysis of 20 prospective cohorts associated the highest dietary linoleic-acid category with colorectal cancer (RR 1.15, 95% CI 1.05–1.27), but found no association for tissue linoleic acid (RR 0.93, 0.61–1.41) or total omega-6 and arachidonic acid.⁷² A broader 2025 umbrella review found generally lower aggregate cancer incidence with higher omega-6 exposure but heterogeneity by site, including signals in opposite directions for different cancers.⁷³ These discordances are compatible with residual dietary confounding, measurement error, multiplicity, or genuine site-specific effects. They justify further research, not a general carcinogenic label.

10 Ultra-processed foods and historical trans fats confound the debate

Seed oils are common ingredients in ultra-processed foods, but ingredient prevalence does not identify the causal feature of the food pattern. Reviews link ultra-processed diets with obesity, but candidate mechanisms include eating rate, texture, energy density, protein leverage, palatability, and food structure—not seed oil alone.⁷⁴ Blaming the shared oil ingredient commits a component-cause error.

Historical terminology creates another category error. “Vegetable oil” once often meant partially hydrogenated shortenings and margarines containing industrial trans fatty acids. Prospective syntheses associate total trans fat with higher all-cause and coronary mortality, and international surveys document substantial declines in industrial trans-fat intake since the mid-1990s.^{75,76,77} Old trials using period margarines therefore cannot be mapped directly onto current nonhydrogenated liquid oils.

Whole foods add a final distinction. Nuts deliver linoleic acid together with fibre, protein, minerals, and a structured

food matrix. Controlled feeding shows benefits from both walnut fatty acids and the whole walnut, with some vascular effects beyond matched oil composition.¹⁵ Genetic analyses show that FADS1 variation changes circulating fatty-acid responses to nut intake without clearly changing diabetes risk.⁷⁸ Food-matrix evidence does not make purified oils harmful; it shows why nutrient equivalence is not whole-food equivalence.

Olive oil provides a useful non-seed comparator for the same reason. In PREDIMED, higher olive-oil intake was associated with lower cardiovascular risk and mortality, while an earlier randomized Mediterranean-diet trial improved blood pressure, glucose, lipid ratios, and C-reactive protein.^{79,80} Those benefits cannot be assigned to oleic acid alone because the intervention changed a dietary pattern and extra-virgin oil carries phenolic compounds. The comparison reinforces the rule: a food source and its matrix can matter without turning linoleic-acid-rich oils into poisons.

11 Population, dose, and genetics modify effects more than the category label

The evidence is strongest for adults replacing saturated fat within an energy-balanced diet. It is weaker for very high linoleic-acid doses, people with established coronary disease, repeated industrial frying exposure, and long-term

cancer endpoints. The old Sydney trial raises particular uncertainty for secondary prevention, while observational benefits are less evident in some cohorts with pre-existing disease.^{38,73} In people with type 2 diabetes, prospective replacement analyses favor PUFA, but intervention reviews remain mixed.^{31,45}

Genetic variation in FADS1 changes how rapidly precursor omega-6 fatty acids are converted to longer-chain metabolites and can alter biomarker responses.^{58,59,78} Current studies do not justify consumer genetic testing or a categorical avoidance recommendation. They instead explain why a single biomarker response may vary between individuals while average clinical effects remain small.

12 Synthesis and practical interpretation

The evidence boundaries are summarized in Figure 1. The main causal contrast is replacement: unsaturated seed oils generally improve LDL cholesterol and probably reduce cardiovascular events modestly when they replace saturated fat; they do not show a general inflammatory or diabetic harm signal. Adding oil without replacing calories, repeatedly reusing frying oil, or treating ultra-processed foods as equivalent to bottled oil asks a different question.

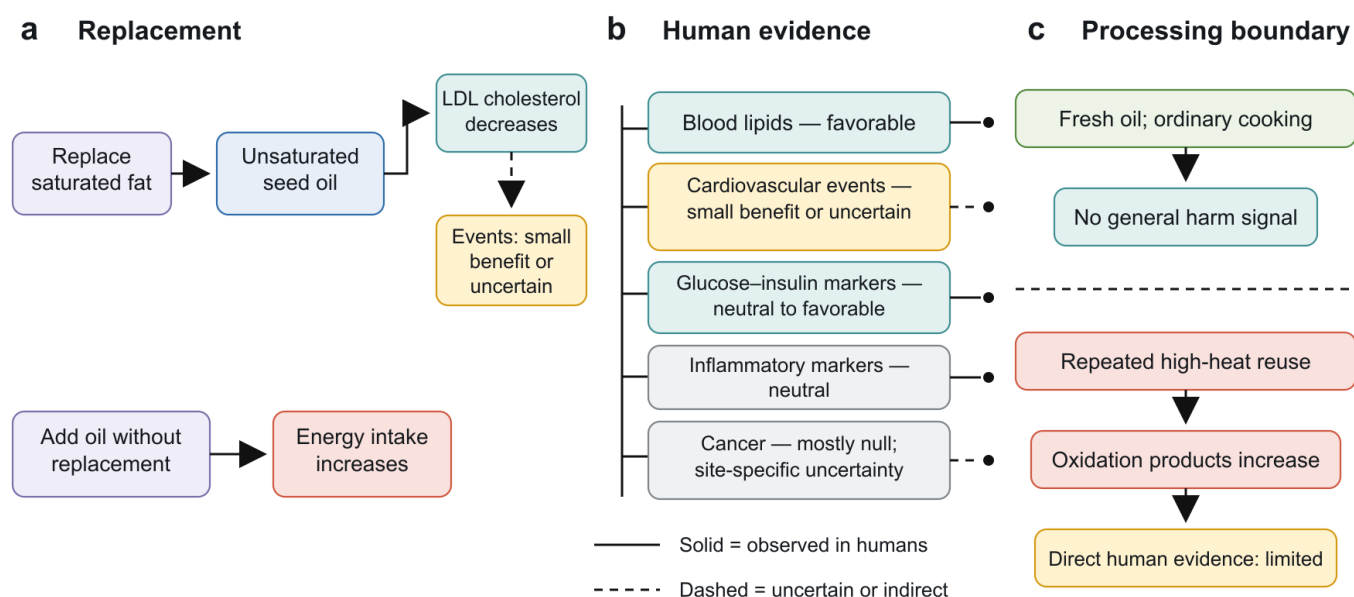


Figure 1. The health effect depends on replacement and processing, not the label “seed oil.” Panel a separates replacing saturated fat from adding energy. Panel b shows strongest human findings: lower LDL cholesterol, small or uncertain event effects, neutral inflammatory markers, and neutral-to-favorable glycaemic markers. Panel c marks repeated high-heat reuse as a distinct oxidation exposure with limited direct human evidence.^{1,20,53,39,62}

For routine use, the evidence supports several practical conclusions. Choose liquid oils rich in unsaturated fat over butter, lard, coconut oil, or palm oil when the culinary role is comparable. Do not add large amounts merely because an oil has a favorable fatty-acid profile; every oil is energy-dense. Avoid repeatedly heating and reusing frying oil, and limit frequent deep-fried foods. Evaluate packaged foods by the whole product—energy density, refined starch, sodium,

fibre, protein, and processing—not by the presence of soy-bean or canola oil alone. People with specific allergies, lipid disorders, gastrointestinal intolerance, or clinician-directed therapeutic diets require individualized advice.

13 Limits of the evidence and research priorities

No long-term randomized trial cleanly compares modern nonhydrogenated seed oils with specific alternatives while

controlling energy intake and measuring major disease outcomes for years. Existing event trials are old and heterogeneous; modern trials are usually short and biomarker-focused; cohorts remain vulnerable to healthy-user effects, dietary measurement error, and uncertain substitution models. Umbrella reviews themselves often inherit overlapping studies and low-quality meta-analyses. ^{1,42,20}

The most useful future studies would randomize well-characterized modern oils against explicit comparators; measure actual fatty-acid intake, oxidation products, and adherence; separate fresh use from repeated frying; prespecify cardiovascular, diabetes, liver, and site-specific cancer outcomes; and include sufficient duration for clinical events. Trials should report the full food pattern and energy balance, because oil effects cannot be interpreted when the displaced nutrient is unknown. Genotype-stratified analyses may be informative, but only if powered for clinical outcomes rather than post hoc biomarker differences.

14 Conclusion

Seed oils are not “really bad” as a general category. The strongest evidence supports a conditional answer: nonhydrogenated oils rich in unsaturated fats are usually favorable when they replace saturated fats, neutral to modestly favorable for glucose–insulin regulation, and not generally proinflammatory in human trials. Prospective cardiovascular and mortality data are mostly favorable or null. The principal counterevidence—two recovered historical trials—creates legitimate uncertainty about large linoleic-acid interventions and mortality but is too era-specific and methodologically constrained to establish modern seed-oil toxicity. Cancer evidence is mostly null, with unresolved site-specific signals. Repeated frying, excessive added energy, industrial trans fats, and ultra-processed foods are genuine concerns, but they should not be conflated with ordinary use of fresh seed oils.

The cross-cutting pattern is consistent across outcomes: the comparator, dose, food matrix, and processing history explain more than the seed origin. A claim that omits those four variables is not yet a scientifically testable claim.

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